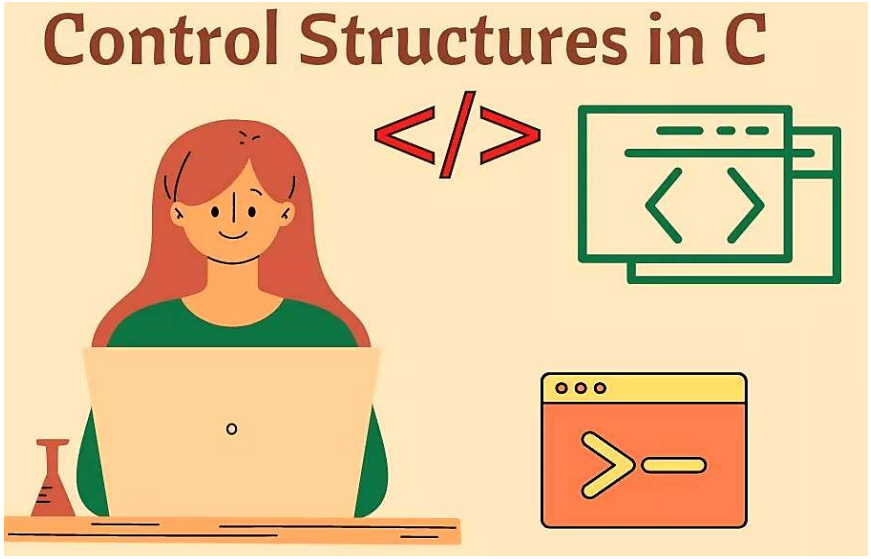
Structured Programming Concepts

(26CSE0105)



Iterative Statements in C

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Topics to be covered

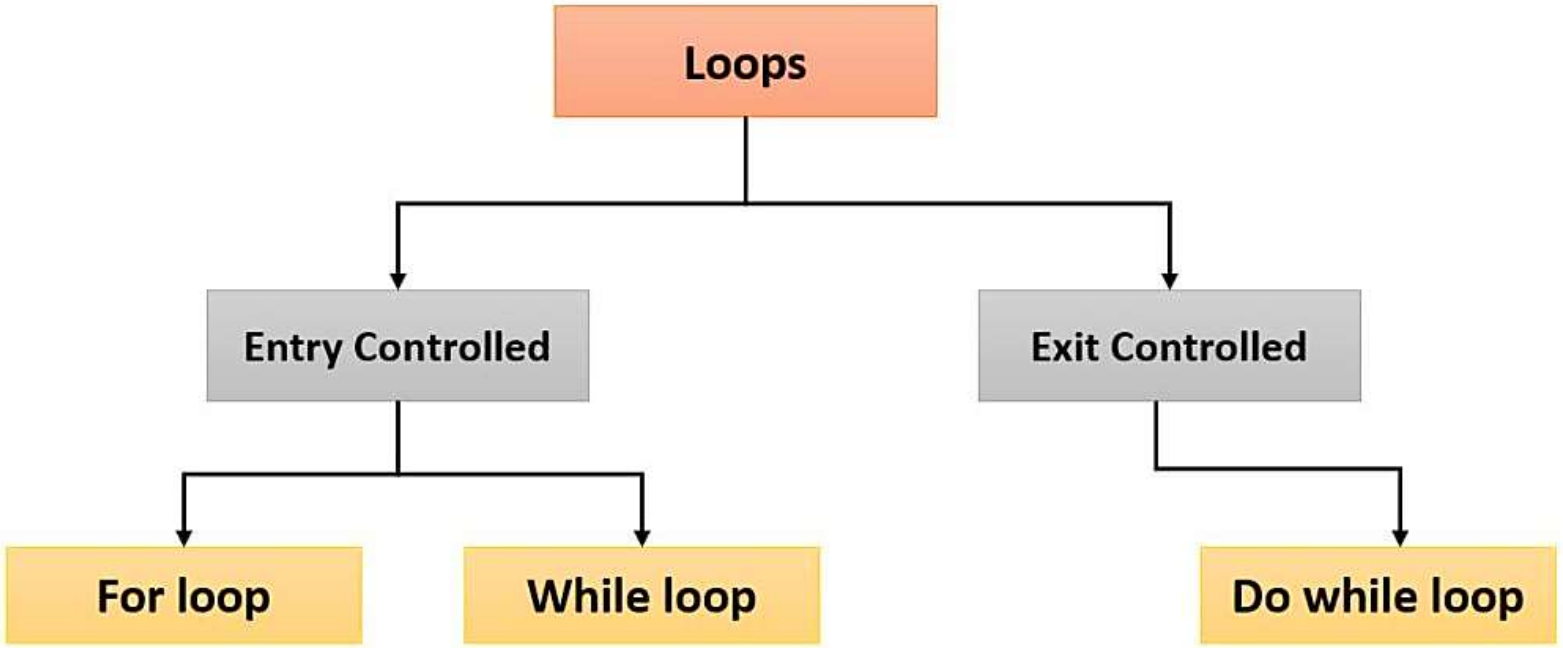
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- ***Iterative Statements (Loop Control):***
  - ***For loop***
  - ***While loop***
  - ***Do-while loops***
- ***Jump Statements:***
  - ***Break***
  - ***Continue***
  - ***goto***

Loop Control Statements

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- ***Repetitive tasks*** *are common in programming, and* ***loops are essential*** *to* ***save time*** *and* ***minimize errors****.*
- ***Used to repeat a block of code*** *until the* ***specified condition is met****.*
- ***Allows*** *programmers* ***to execute*** *a statement or group of statements* ***multiple times*** *without writing the* ***code again and again****.*

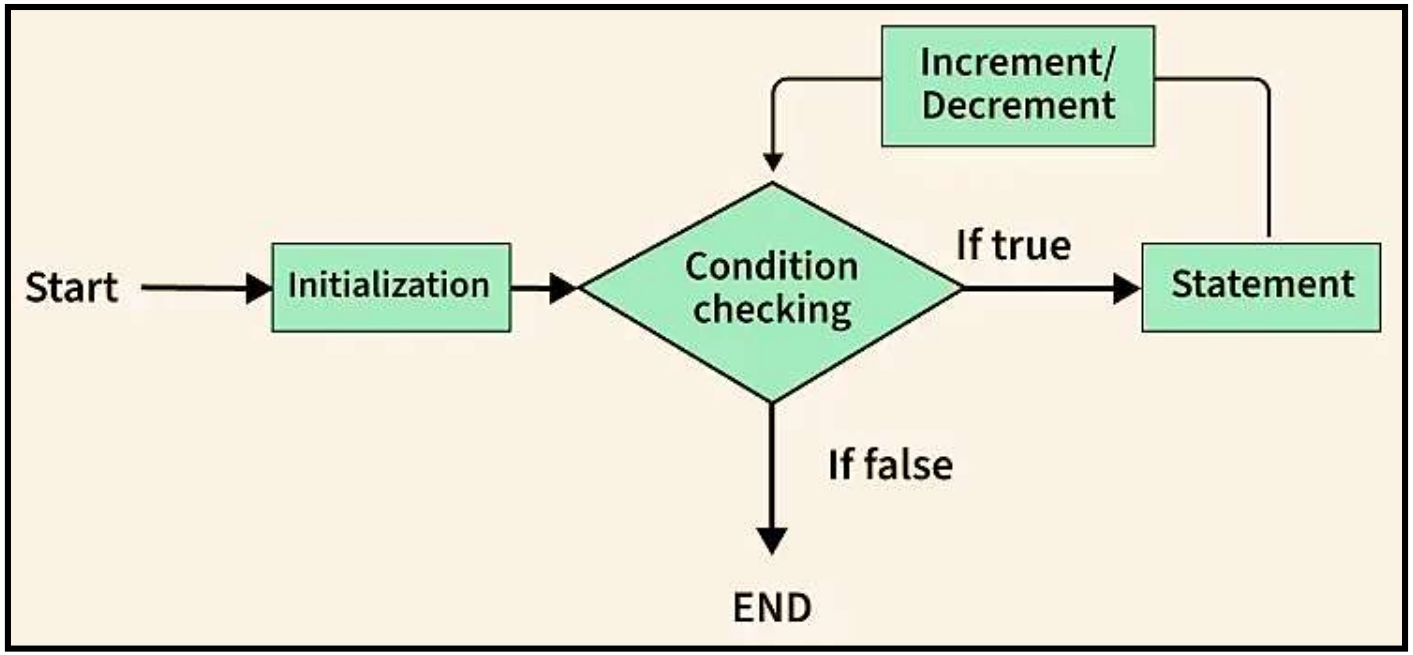


For Loop

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- *The **for loop** is an **entry-controlled loop** that executes the statements **till the given condition is true**.*

```
for (init; condition; increment)
{
    statement(s);
}
```



For Loop (example)

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```
#include <stdio.h>
int main()
{
  int a;
  // for loop execution
  for(a = 1; a <= 5; a++)
  {
    printf("a: %d\n", a);
  }
  return 0;
}
```

```
a: 1
a: 2
a: 3
a: 4
a: 5

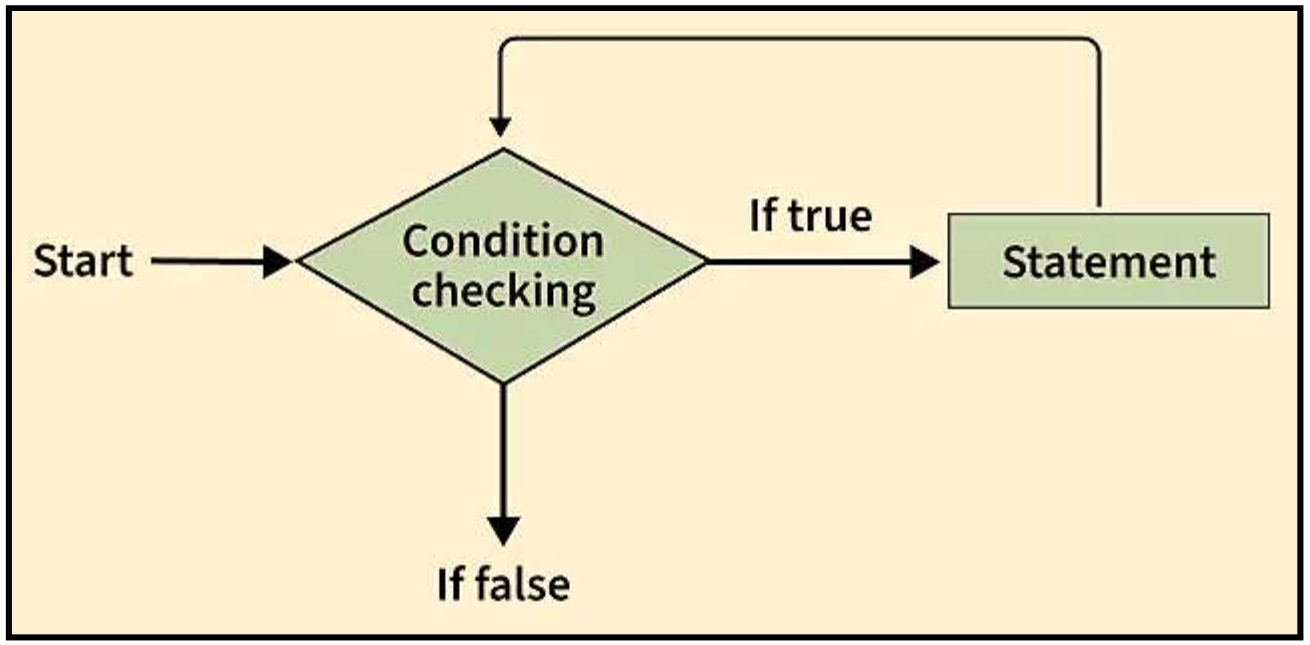

...Program finished with exit code 0
Press ENTER to exit console.
```

While Loop

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- *The **while loop** is also an **entry-controlled loop**.*
- *Used to repeat a block of code an **unknown number of times** until a **specific condition is met**.*

```
while(expression)
{
    statement(s);
}
```



While Loop (example)

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```
#include <stdio.h>
int main()
{
  int a = 1;
  while(a <= 5)
  {
    printf("Hello World \n");
    a++;
  }
  return 0;
}
```

```
Hello World
Hello World
Hello World
Hello World
Hello World

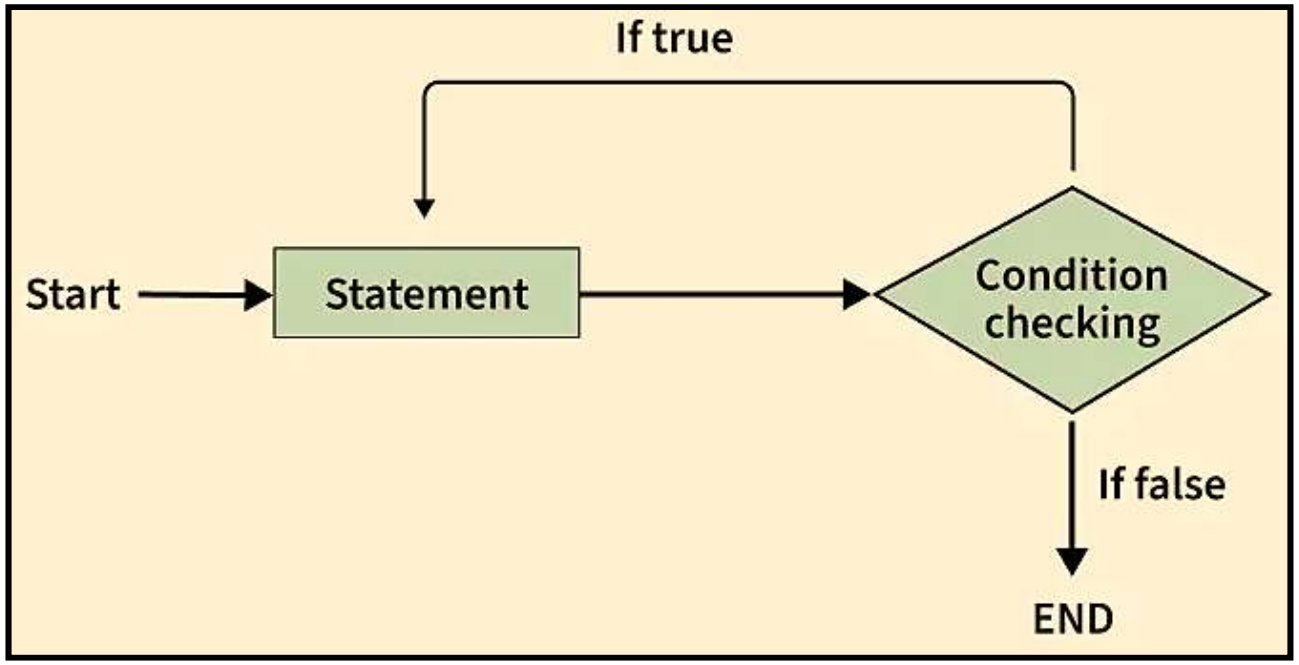

...Program finished with exit code 0
Press ENTER to exit console.
```

Do-while Loop

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- *The **do-while** is an **exit-controlled loop** where the **condition is checked** after **executing the body of loop**.*

```
do
{
   statement(s);
} while(condition);
```



Do-while Loop (example)

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```
#include <stdio.h>
int main()
{
  int a = 1;
  do
  {
    printf("Hello World\n");
    a++;
  } while(a <= 5);
  return 0;
}
```

```
Hello World
Hello World
Hello World
Hello World
Hello World


...Program finished with exit code 0
Press ENTER to exit console.
```

Difference b/w for & while loop

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For Loop	While Loop
The for loop is used for definite loops when the number of iterations is known.	The while loop is used when the number of iterations is not known.
For loops can have their counter variables declared in the declaration itself.	There is no built-in loop control variable with a while loop.
The loop iterates infinite number of times if the condition is not specified.	If the condition is not specified, it shows a compilation error.

Difference b/w while & do-while

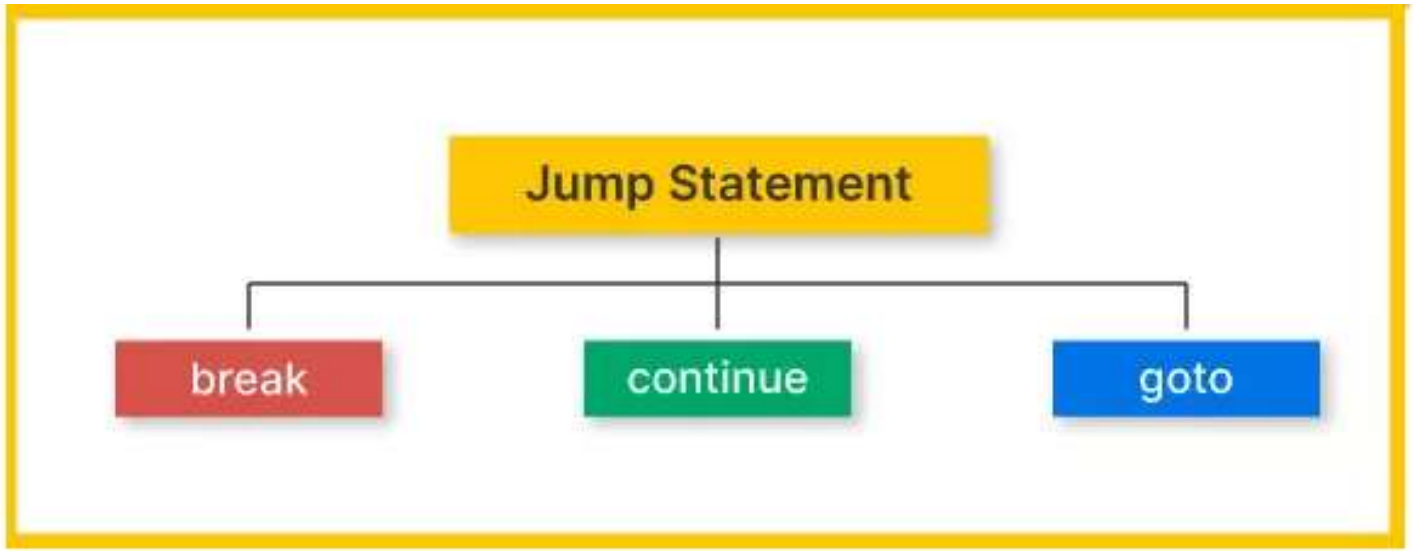
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While Loop	Do-While Loop
The while loop evaluates the condition first and then execute the statements.	The do-while loop executes the statements first before evaluating the condition.
The condition is specified at the beginning of the loop.	The condition isn't specified until after the body of the loop.
The body is executed only if a certain condition is met and it terminates when the condition is false.	The body is always executed at least once, regardless of whether the condition is met.

Jump Statements in C

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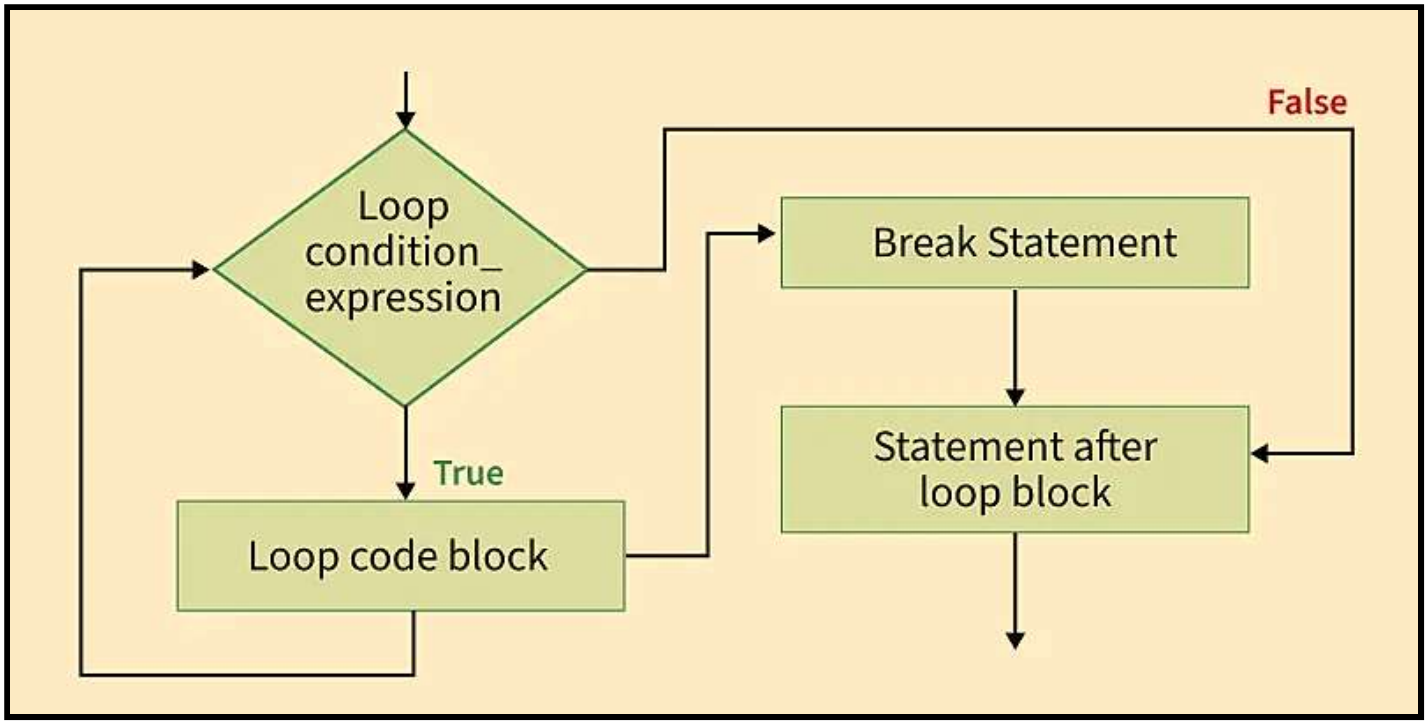
- *Used to jump from **one part** of code to **another altering the normal flow** of the program.*
- *Used to **transfer program control** to **somewhere else** in the program.*



Break Statement in C

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- ***Terminates the loop*** *or* ***switch*** *statement* ***prematurely****, before the* ***loop*** *or* ***switch*** *block has been* ***fully executed****.*



Break Statement (example)

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```
#include <stdio.h>
int main()
{
    int i;
    for (i = 1; i <= 10; i++)
    {
        if (i == 6)
        {
            break;
        }
        printf("%d ", i);
    }
    printf("Loop exited due to break.\n");
    return 0;
}
```

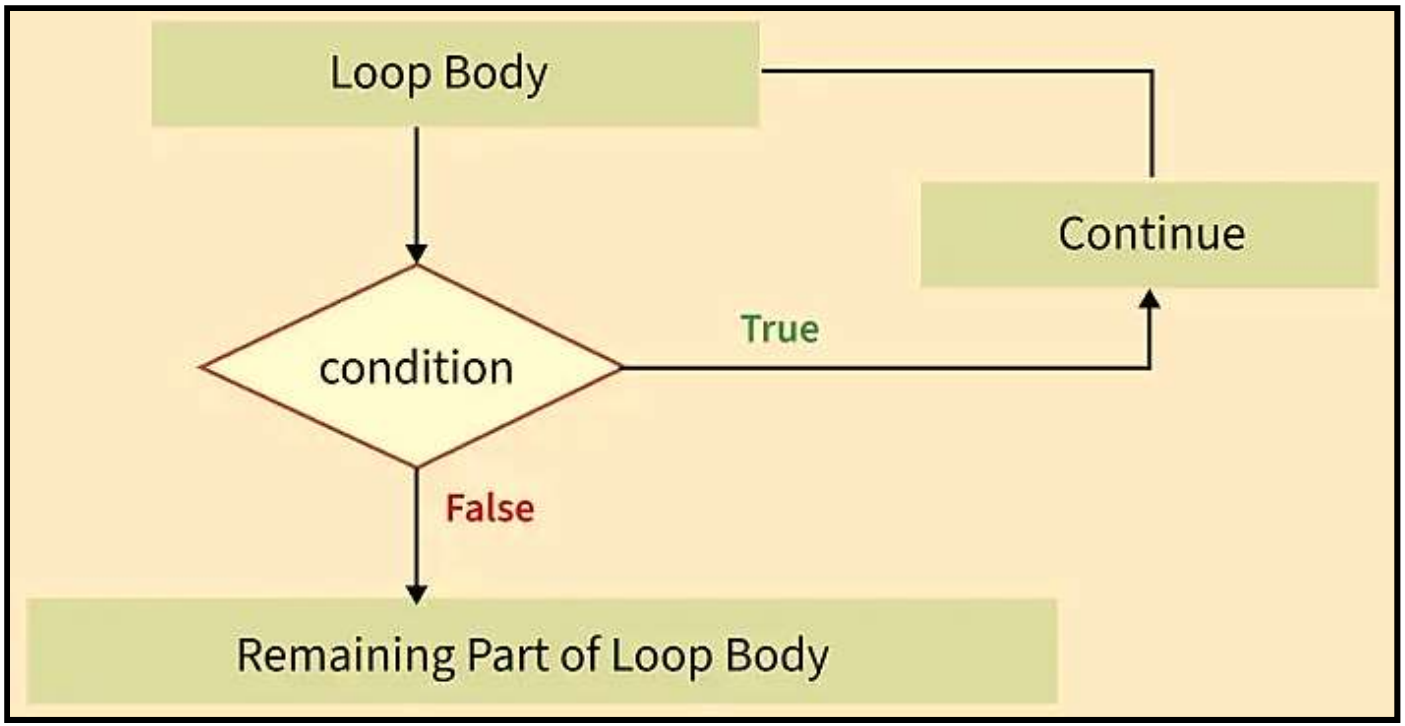
```
1 2 3 4 5 Loop exited due to break.


...Program finished with exit code 0
Press ENTER to exit console.
```

Continue Statement in C

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- *Used to **skip** the **current iteration** and **jump** to the **next iteration** of the **loop**.*



Continue Statement (example)

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```
#include <stdio.h>
int main()
{
    int i;
    for (i = 1; i <= 5; i++)
    {
        if (i == 2)
        {
            continue;
        }
        printf("Executing iteration %d\n", i);
    }
    return 0;
}
```

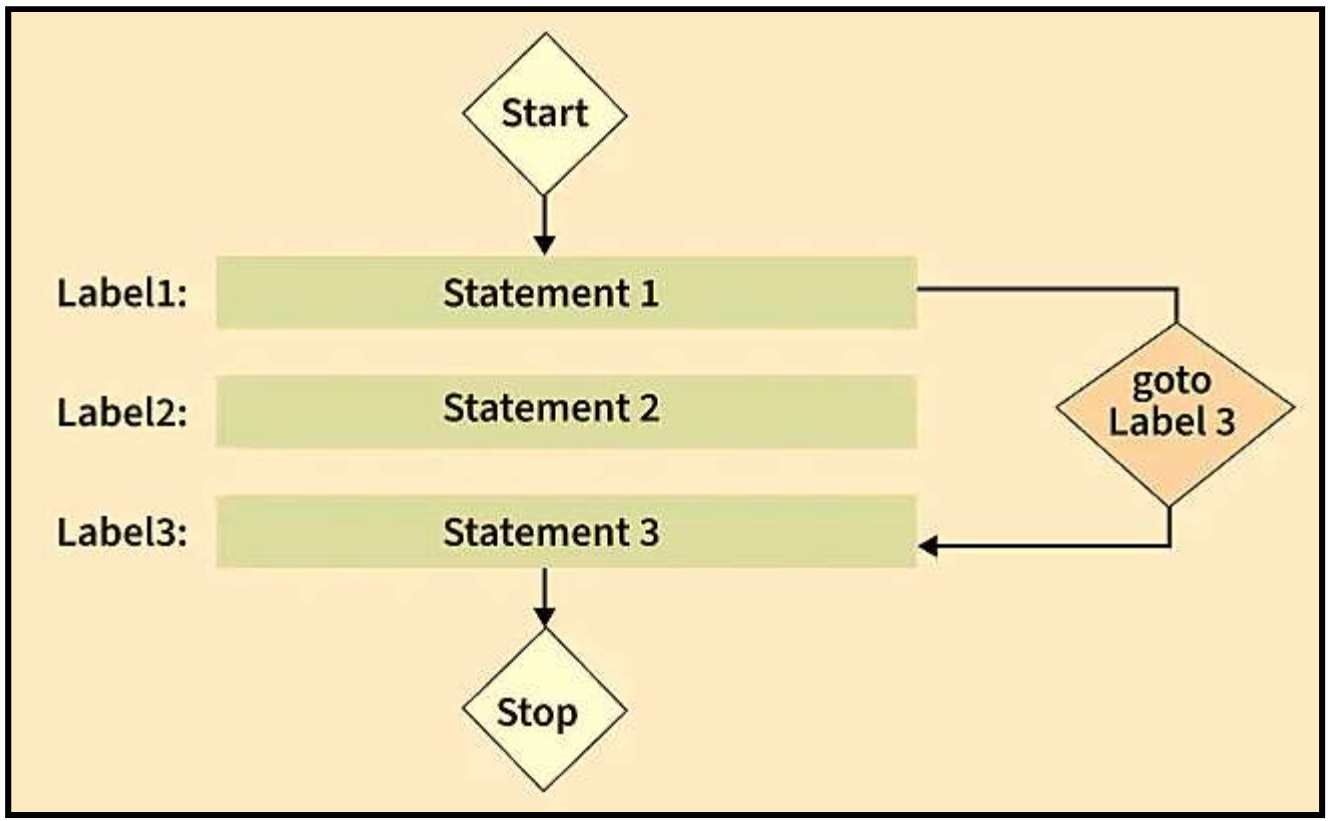
```
Executing iteration 1
Executing iteration 3
Executing iteration 4
Executing iteration 5


...Program finished with exit code 0
Press ENTER to exit console.
```

goto Statement in C

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- *Used to **jump to a certain location** within a **function** from **any other location**. The **program control** is moved to a **labeled statement** inside the **same function**.*



goto Statement (example)

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```
#include <stdio.h>
int checkEvenOdd(int num)
{
if (num % 2 == 0)
        goto even;
else
        goto odd;
even:
    printf("Entered number %d is even.", num);
    return 0;
odd:
   printf("Entered number %d is odd.", num);
   return 0;
}
```

```
int main()
{
int num = 21;
checkEvenOdd(num);
return 0;
}
```

```
Entered number 21 is odd.

...Program finished with exit code 0
Press ENTER to exit console.
```

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